

CAIRNS AAF, AL, USA

WMO: 722269

Lat: 31.267N

Lon: 85.717W

Elev: 92

StdP: 100.23

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03850

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.0	-1.3	-12.1	1.3	1.4	-9.1	1.8	3.5	9.2	12.2	8.4	11.8	2.7	320	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	35.4	24.4	34.1	24.5	33.0	24.3	26.9	31.5	26.3	31.0	25.7	30.3	3.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.6	28.6	25.1	20.5	27.9	24.7	19.9	27.5	84.8	31.6	82.0	31.0	79.6	30.9	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	7.0	5.8	DB	-6.8	36.9	2.3	1.3	-8.5	37.8	-9.9	38.6	-11.2	39.4	-12.9	40.3	(4)
(5)			WB	-7.9	28.1	2.3	1.1	-9.6	28.8	-10.9	29.5	-12.3	30.1	-14.0	30.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.4	9.9	12.1	15.5	19.0	23.5	26.5	27.5	27.0	25.0	20.0	14.5	11.6	(6)
(7)		DBStd	7.37	5.71	5.28	4.69	3.53	3.04	2.13	1.59	1.67	2.69	4.07	4.53	5.30	(7)
(8)		HDD10.0	176	74	35	12	0	0	0	0	0	0	1	11	44	(8)
(9)		HDD18.3	977	266	182	111	35	4	0	0	0	2	31	130	216	(9)
(10)		CDD10.0	3598	71	94	181	269	418	496	543	527	449	309	147	94	(10)
(11)		CDD18.3	1357	4	8	21	54	164	246	284	268	201	82	16	8	(11)
(12)		CDH23.3	12313	13	39	169	496	1527	2332	2748	2488	1721	659	94	28	(12)
(13)		CDH26.7	5180	0	2	21	117	621	1098	1284	1117	723	186	10	1	(13)
(14)	Wind	WSAvg	2.7	3.1	3.2	3.2	3.0	2.7	2.4	2.2	2.2	2.5	2.7	2.7	2.9	(14)
(15)	Precipitation	PrecAvg	1407	134	116	141	115	75	140	163	121	108	80	112	102	(15)
(16)		PrecMax	1794	350	240	280	240	225	301	513	208	428	204	306	236	(16)
(17)		PrecMin	925	34	31	9	32	5	53	63	59	19	12	34	38	(17)
(18)		PrecStd	257	71	59	74	58	50	76	98	44	94	50	66	45	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	26.5	28.9	31.0	35.0	37.3	37.1	36.1	35.6	32.2	27.8	26.1	(19)
(20)			MCWB	18.8	19.4	18.9	20.8	22.4	24.1	25.0	24.6	23.3	22.8	20.9	20.7	(20)
(21)		2%	DB	22.7	24.6	27.1	29.3	33.0	35.3	35.0	34.4	33.6	30.3	25.9	23.8	(21)
(22)			MCWB	17.9	18.6	18.3	20.3	22.0	24.2	25.3	25.1	23.2	21.8	19.5	19.5	(22)
(23)		5%	DB	21.1	22.4	25.2	27.9	31.7	33.9	33.8	33.2	32.3	28.8	24.1	22.1	(23)
(24)			MCWB	17.6	17.4	17.6	19.4	21.9	24.3	25.2	24.9	23.5	21.1	18.7	19.0	(24)
(25)		10%	DB	18.9	20.7	23.5	26.4	30.2	32.4	32.6	32.2	30.9	27.2	22.5	20.1	(25)
(26)			MCWB	16.3	16.8	16.8	18.4	21.5	24.0	25.0	24.6	23.2	20.1	18.0	17.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.8	21.2	21.9	23.4	24.8	26.9	27.8	28.0	26.1	25.4	23.3	22.6	(27)
(28)			MCDWB	22.1	23.8	25.0	27.0	29.8	32.2	32.5	32.2	30.0	28.5	25.1	24.2	(28)
(29)		2%	WB	19.6	20.2	20.7	22.3	24.0	26.1	27.0	27.0	25.5	24.0	21.9	20.8	(29)
(30)			MCDWB	21.6	22.6	23.4	26.1	28.7	31.5	31.8	31.1	29.7	26.9	23.7	22.3	(30)
(31)		5%	WB	18.3	19.0	19.7	21.4	23.5	25.5	26.3	26.2	24.9	23.0	20.5	19.6	(31)
(32)			MCDWB	20.2	21.6	23.0	25.1	28.1	30.8	31.3	30.4	29.2	26.3	22.8	21.3	(32)
(33)		10%	WB	16.8	17.6	18.6	20.4	22.8	25.0	25.8	25.6	24.3	22.1	19.0	18.2	(33)
(34)			MCDWB	18.8	19.6	21.9	24.0	27.5	30.0	30.4	29.7	28.4	25.3	21.7	19.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.8	12.1	12.7	13.0	12.3	10.9	10.4	10.3	10.8	12.3	12.8	11.1	(35)
(36)			MCDBR	11.9	12.7	13.7	14.0	13.6	13.0	11.9	11.8	12.5	13.0	13.0	11.5	(36)
(37)			MCWBR	7.8	7.6	6.7	6.0	4.5	3.9	3.9	4.0	4.3	5.5	7.6	7.8	(37)
(38)		5% WB	MCDWB	10.5	10.6	11.1	11.0	11.4	11.2	10.6	10.0	10.0	9.8	10.6	10.1	(38)
(39)			MCWBR	8.1	7.6	6.6	6.0	4.6	4.2	4.2	4.0	4.0	5.1	7.5	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.324	0.333	0.356	0.381	0.433	0.469	0.513	0.501	0.431	0.372	0.345	0.330	(40)	
(41)		taud	2.521	2.497	2.433	2.368	2.256	2.215	2.104	2.136	2.310	2.449	2.470	2.519	(41)	
(42)		Ebn at Noon	905	929	926	910	861	825	787	793	839	873	871	878	(42)	
(43)		Edh at Noon	86	97	111	123	139	144	160	153	124	101	90	83	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.99	3.58	4.79	5.86	6.41	6.10	5.93	5.51	5.00	4.35	3.37	2.60	(44)	
(45)		RadStd	0.19	0.45	0.39	0.48	0.44	0.48	0.37	0.37	0.37	0.48	0.29	0.28	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (44 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	-93	+166	+109	(47)

Nomenclature: See separate page